Overview Data Analytics and R

Overview

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Data Analytics–Definition

- Data analytics (BA) refers to the tools, techniques and processes for continuous exploration and investigation of past data to gain insights and help in decision making.

- Data Analytics is an integration between science, technology and business context that assist data driven decision making.

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Data Explosion

- About 350 million photos are uploaded every day in the Facebook
- Amount of credit card debt in US: $762.1 billion
- Amount of credit card debt in India: Rs. 45,383 crore ($709 million)
- Loss due to global Credit card and debit card fraud $21.84 billion during 2015
- Every day, Walmart processes $36 million dollars an hour in sales
- BMTC with approx. 6000 buses plying in Bangalore sends 1 billion signals to the server updating its location every month

Interesting Stats: http://expandedramblings.com/

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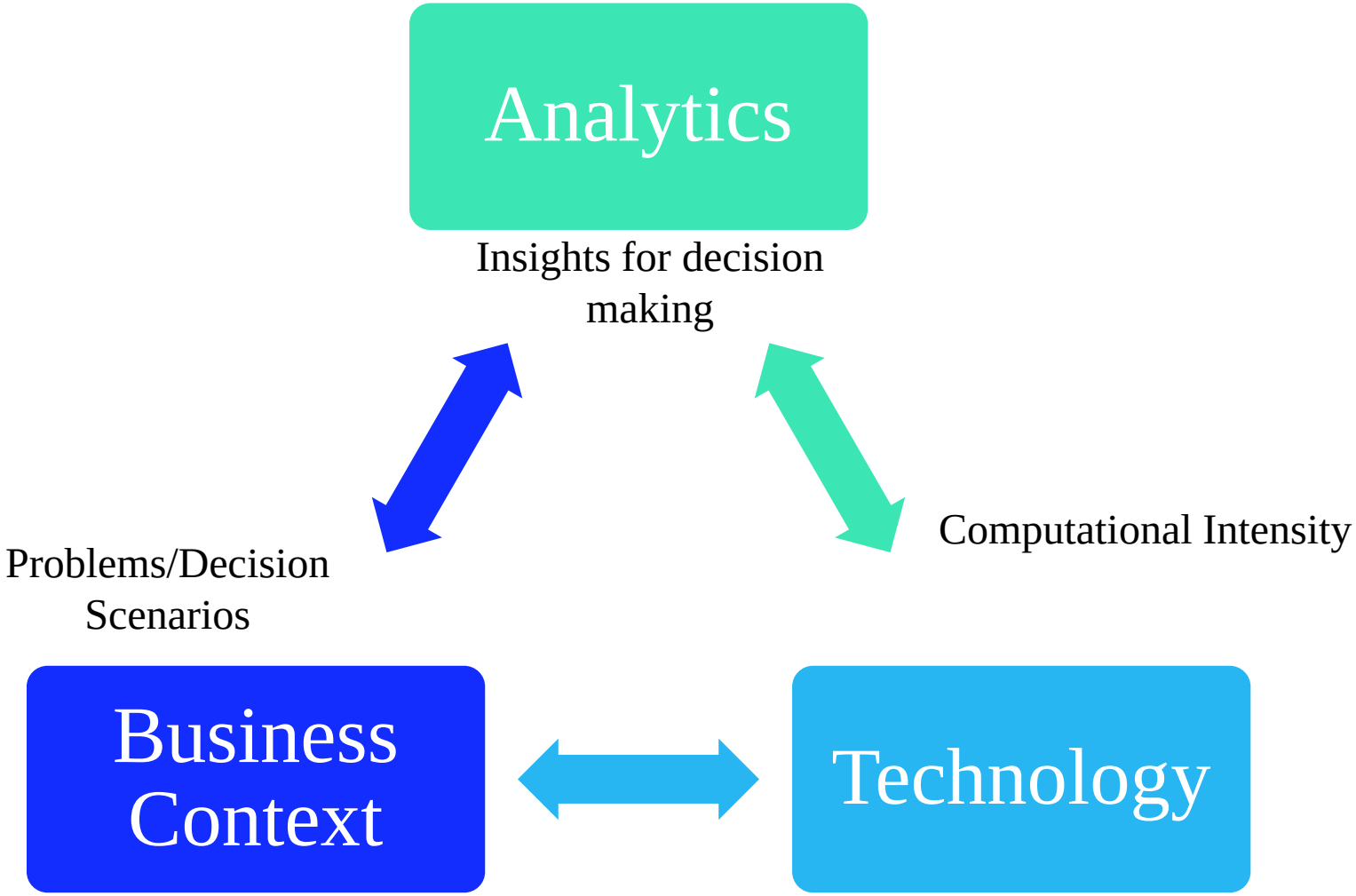
Analytics Trilogy



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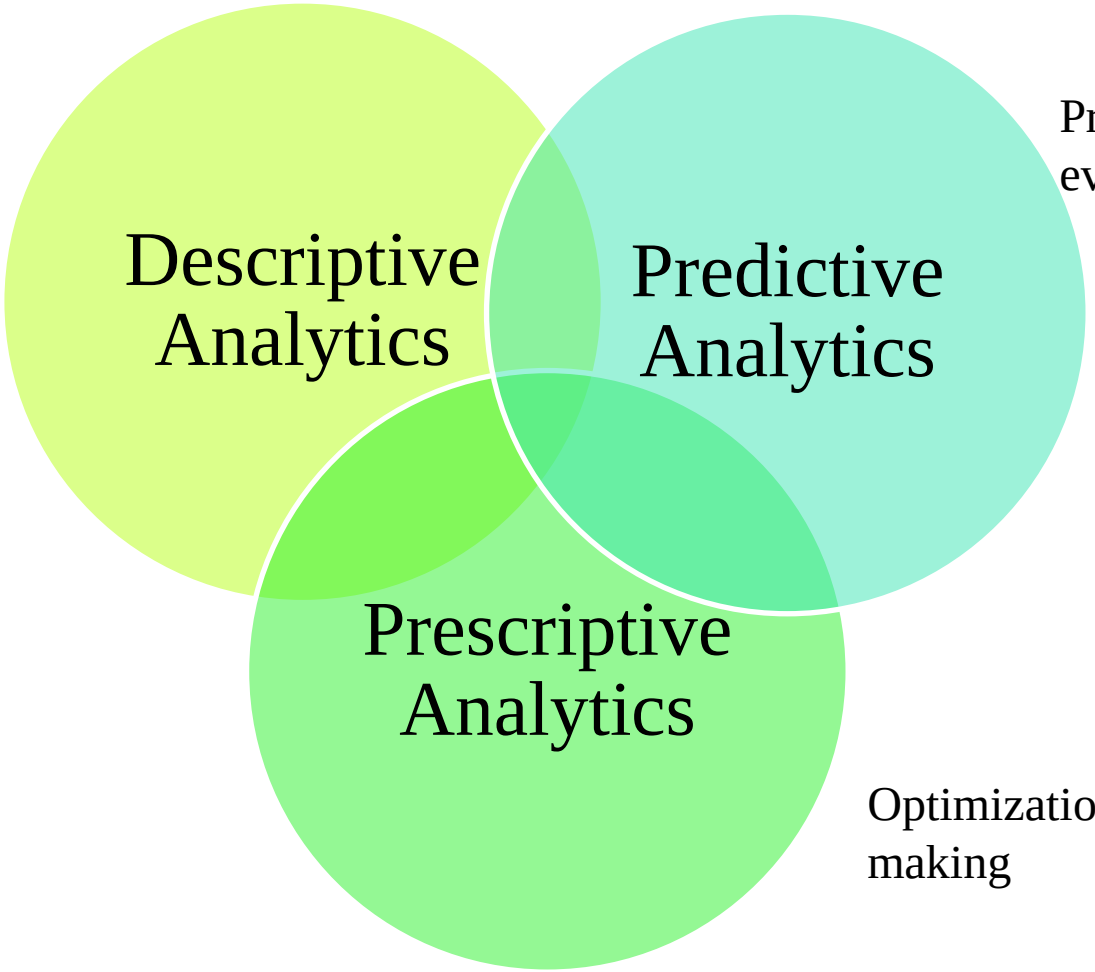
Components of Analytics

Data synthesis and Visualization



Predicting future events

Optimization and decision making

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Components of Analytics

Knowing what happened in past and what may happen in future, what optimal strategy can be adopted to achieve an objective like maximize profit.

Prescriptive

Learning from past data and predicting what may happen in future and likelihood of happening in future.

Predictive

Optimal product pricing or product mix strategies.

Understanding what happened and why happened by exploring past data.

Descriptive

Product sales or revenue forecast.

Product sales patterns or factors influencing product sales.

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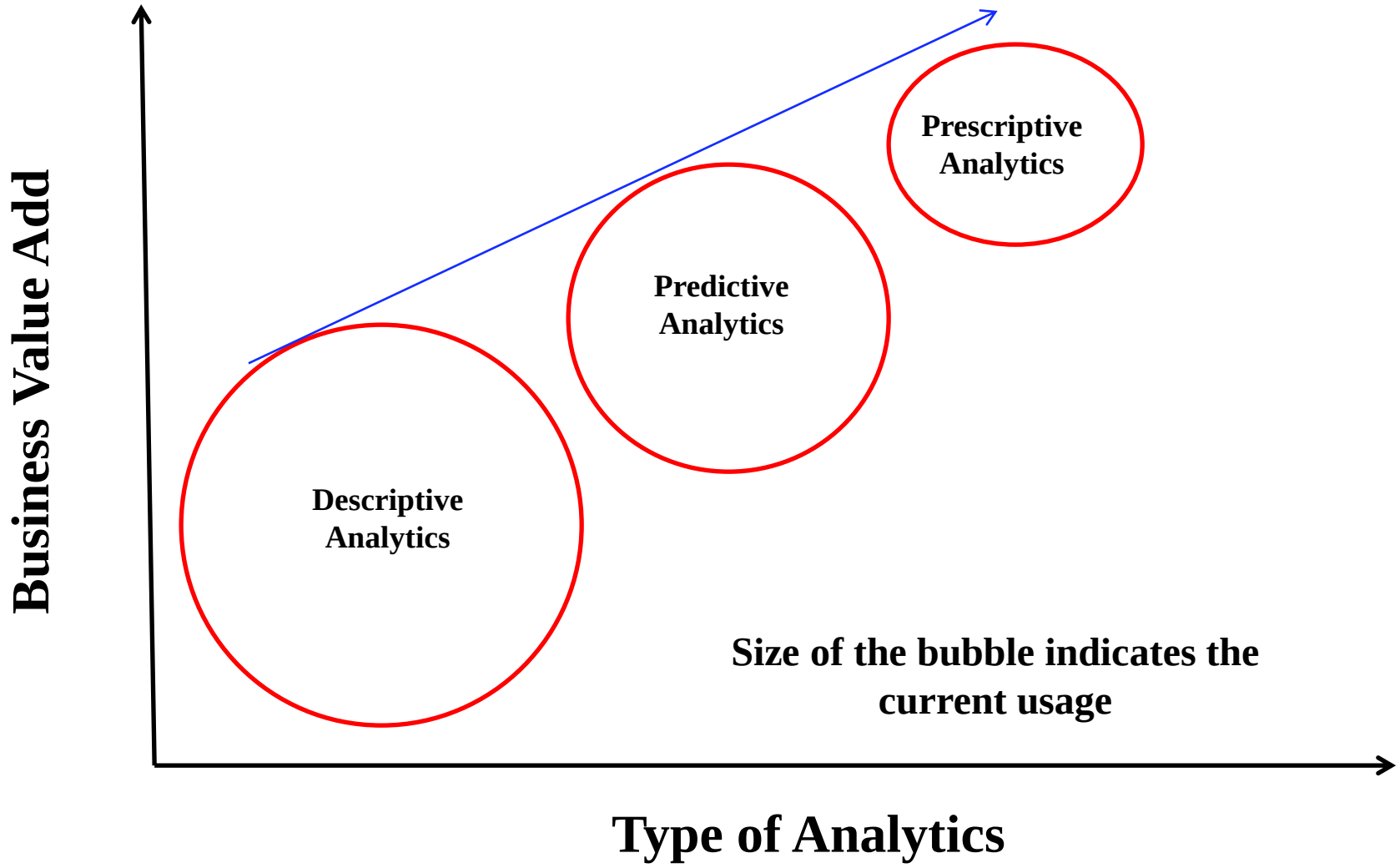

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Power of Descriptive Analytics

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Descriptive Analytics Applications

- Most shoppers turn towards right when they enter the a retail store.
- Conversion rate of women shoppers is higher than male shoppers among electronic gadgets purchasers (Radio Shack).
- Strawberry pop-tarts sell 7 times more during hurricane compared to regular period (Wal Mart).
- Women car buyers prefer women sales person.

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Predictive Analytics Application

- Which product the customer is likely to buy in his next purchase (recommender system).
- Which customer is likely to default in his/her loan payment.
- Who is likely to cancel the product that was ordered through e-commerce portal.

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Prescriptive Analytics Application

- What is the optimal route for a delivery truck.
- Whether a company should introduce a new product?
- What is the optimal product mix?
- How to manage the fleet of vehicles owned by a company for employee drop and pick up?

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R Vs. Python

R	Python
Built for Statistical Analysis.	General Purpose Language. Main objective is productivity and readability.
Primarily used in academics and research. Enterprise have started adopting it for analysis.	Has a very strong presence in enterprises for large number of software developments. Easier adoption in enterprises as strong development experience already exists.
Integration with other enterprise systems are not straightforward.	Integration with other enterprise systems or applications are easier.

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Overview and History of R

R is an implementation of the S programming language combined with lexical scoping semantics inspired by Scheme (S language).

- S was created by John Chambers in 1976, while at Bell Labs.
- R was created by **Ross Ihaka** and **Robert Gentleman** at the University of Auckland, New Zealand.
- The project was conceived in 1992, with an initial version released in 1995 and a stable beta version in 2000.
- currently being developed by the R Development Core Team, of which John Chambers is a member.

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Overview and History of R

Features:

- Runs on almost any standard computing platform/OS
- Frequent releases (annual + bug fix releases); active development
- Quite lean, as far as software goes; functionality is divided into modular packages
- Very sophisticated graphics capabilities; better than most stat packages
- Useful for interactive work, but contains a powerful programming language for developing new tools
- Very active and vibrant user community; contains R-help and R-developer mailing lists and Stack Overflow

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Design of the R System

- The R system is divided into two conceptual parts:

Base R	R Extension
• Base R can be downloaded from CRAN (http://cran.r-project.org). • The base R system contains, among other things, the base package, which is required to run R and contains the most fundamental functions.	• There are about 4000 packages on CRAN that have been developed by users and programmers around the world. • There are also many packages associated with the Bioconductor project (http://bioconductor.org).

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R Resources

- Number of references and reading material can be found on R.

R resources available from CRAN	Standard books
• An Introduction to R • Writing R Extensions • R Data Import/Export • R Installation and Administration (mostly for building R from sources) • R Internals (not for the faint of heart)	• Chambers (2008). Software for Data Analysis, Springer (your textbook) • Chambers (1998). Programming with Data, Springer • Venables & Ripley (2002). Modern Applied Statistics with S, Springer • Venables & Ripley (2000). S Programming, Springer • Pinheiro & Bates (2000). Mixed-Effects Models in S and S-PLUS, Springer • Murrell (2005). R Graphics, Chapman & Hall/CRC Press

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R Dataset and Tools

- R comes with a number of sample datasets that you can experiment with.

On R Studio Console:

```
data( ) #to see the available datasets in the installed packages
help(datasetname)  #for details on a sample dataset
```

Help on the error: "Failed to set default locale". Open terminal and type the following:

```
defaults write org.R-project.R force.LANG en_US.UTF-8
```

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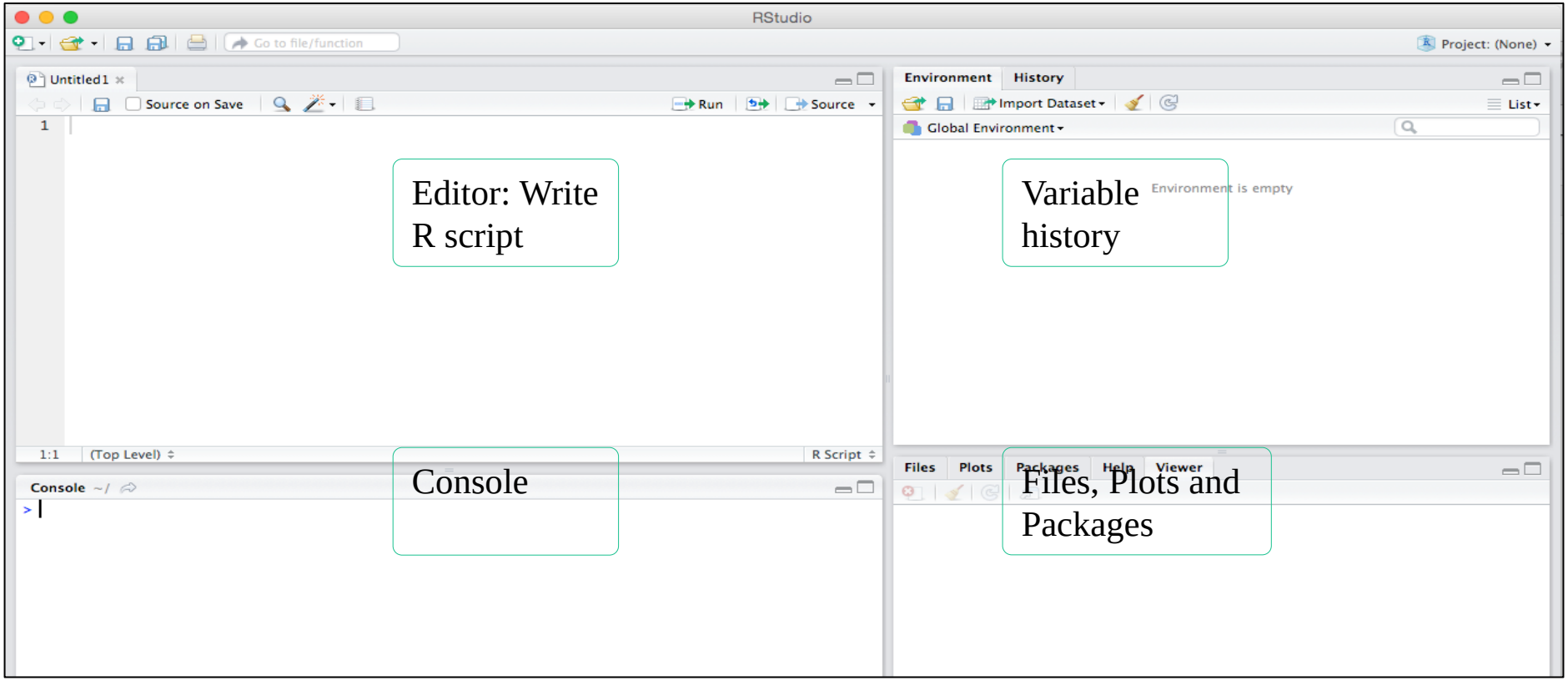
Introducing R Studio

- R Studio (version 0.99.442) console has four primary blocks:



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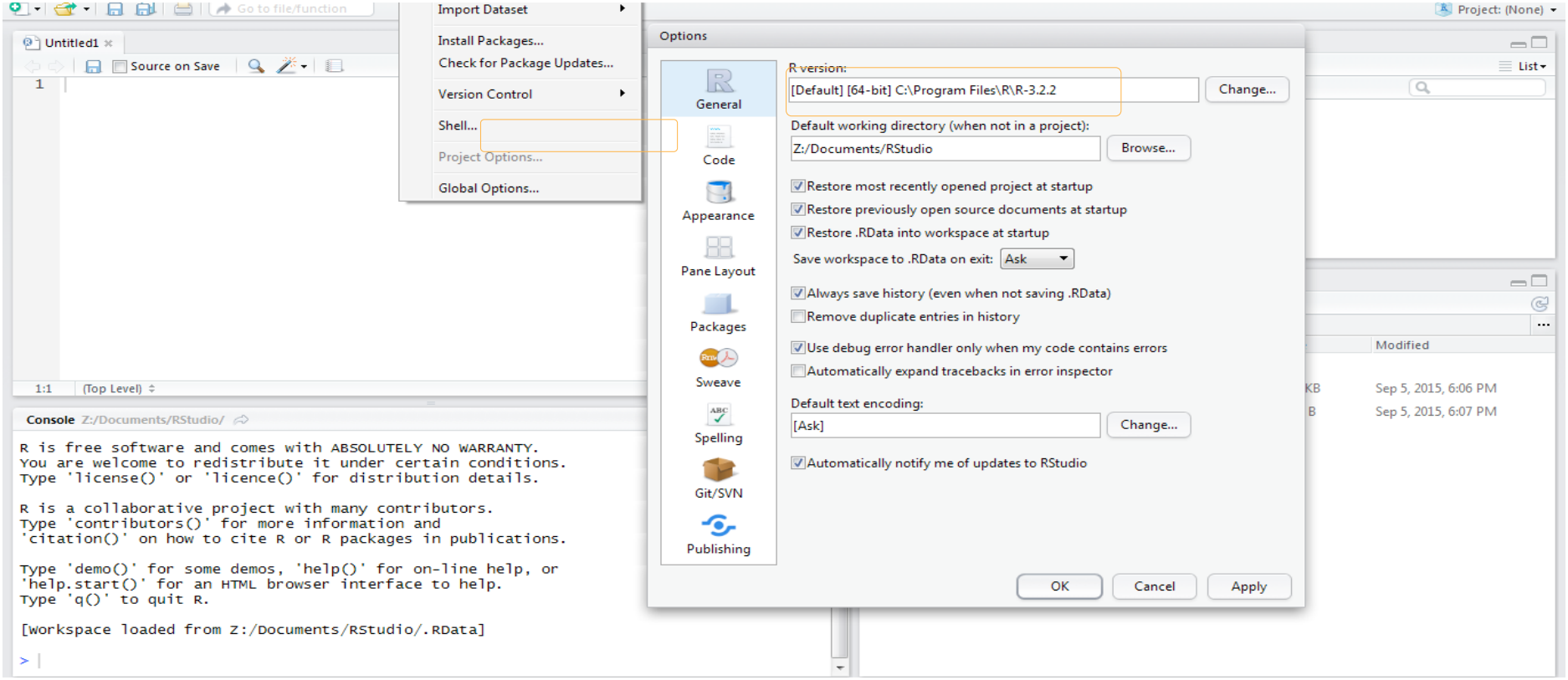
R Studio—Setting Global Working Directory

- Set the global working directory through the option below:



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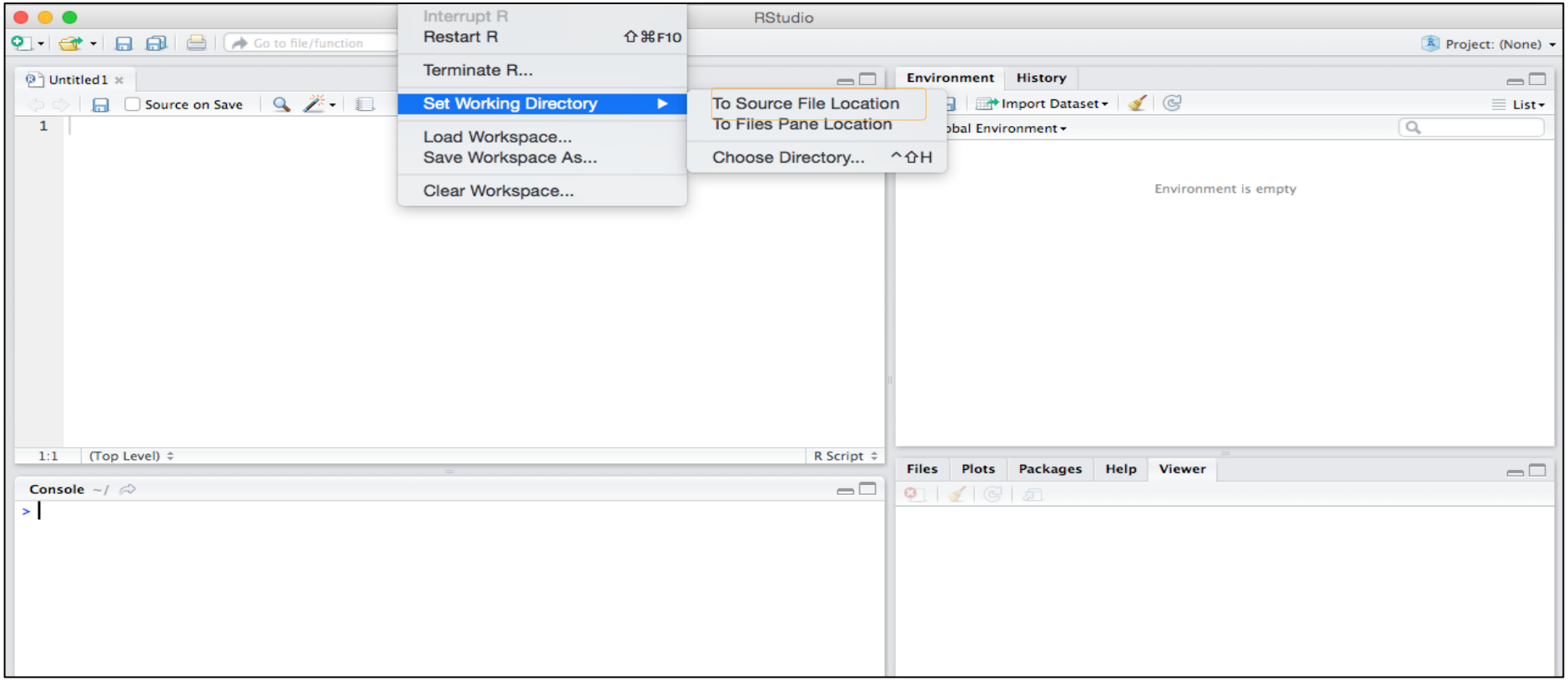
R Studio—Setting Local Working Directory

- Local working directory can be set as shown below. Two useful commands: getwd() and setwd()



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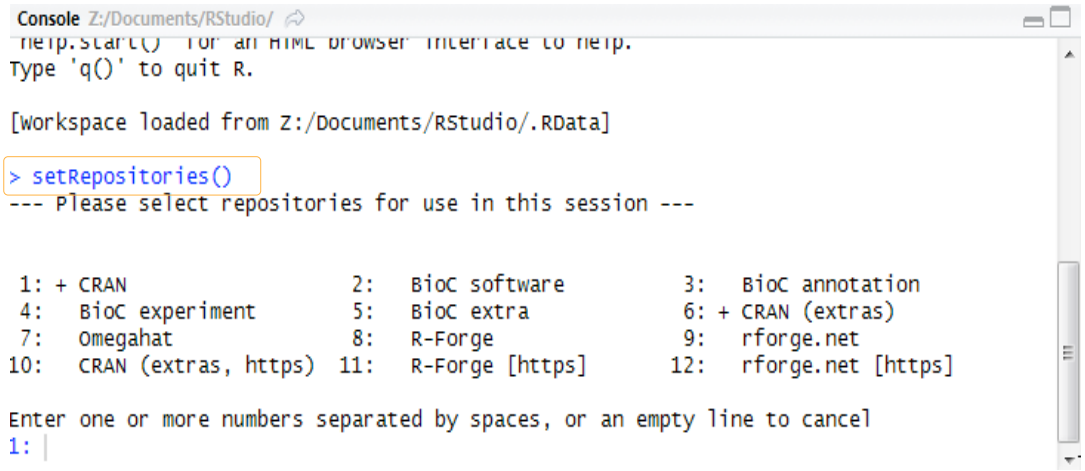
Choosing a CRAN Mirror and Setting Repositories

- CRAN mirrors contain R packages that can extend the functionality of R.
- Choose a mirror located close to you as that will most likely give you the fastest downloads
- Repositories host the packages. Some of the examples of repositories are:

CRAN, CRANextra, BioCsoft, BioCann, BioCexp, BioCext, Omegahat, R-Forge and rforge.net



- Use this code to set your repositories: `setRepositories()`

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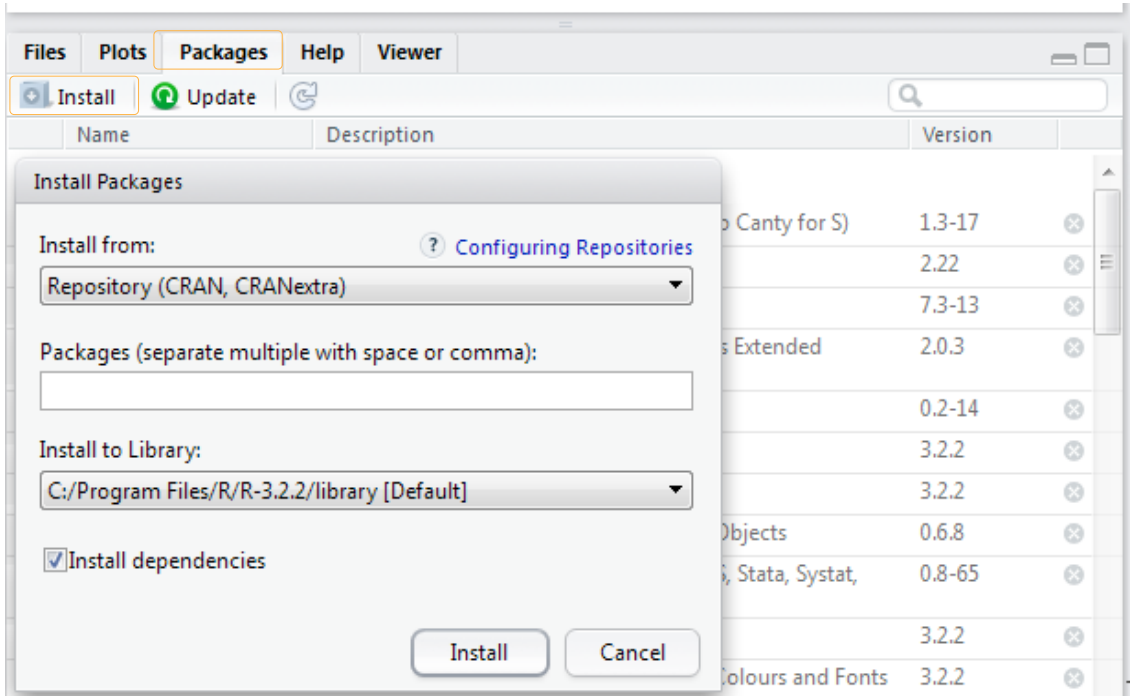
Package TAB

Packages can be installed as and when required.

- The Packages tab lets you see what packages are installed
- A package must be loaded before you can use it
  - In Rstudio, this is accomplished by clicking the checkbox next to the package name in the package tab



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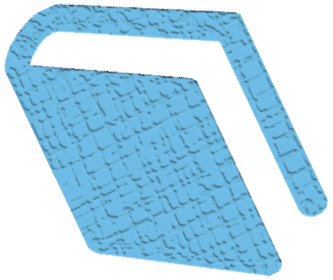
Summary

Summary of the topics covered in this lesson:



- With the data explosion across industry, the usage of analytics in decision making will become the most critical factor for being competitive in business.
- Descriptive analytics becomes the stepping stone to all the complex problems which can be solved using analytics.

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End of Lesson–Overview to Analytics and R

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